

Quantum Computing, Quantum Machine Learning and Quantum Information Theories

Morten Hjorth-Jensen¹

Department of Physics, University of Oslo, Norway¹

February 5, 2025

© 1999-2025, Morten Hjorth-Jensen. Released under CC Attribution-NonCommercial 4.0 license

Plans for the week of February 3-7: Entanglement, entropies and density matrices

1. Reminder and review of density matrices and measurements from last week
2. Schmidt decomposition and entanglement
3. Discussion of entropies, classical information entropy (Shannon entropy) and von Neumann entropy
4. Video of lecture to be added

Chapters 3 and 4 of Scherer's text contains useful discussions of several of these topics. More reading suggestions will be added.

Motivation

In order to study entanglement and why it is so important for quantum computing, we need to introduce some basic measures and useful quantities. For these endeavors, we will use our two-qubit system from the second lecture in order to introduce and repeat, through examples, density matrices and entropy. These two quantities, together with technicalities like the Schmidt decomposition define important quantities in analyzing quantum computing examples.

The Schmidt decomposition is again a linear decomposition which allows us to express a vector in terms of tensor product of two inner product spaces. In quantum information theory and quantum computing it is widely used as a way to define and describe entanglement.

Reminder on density matrices and traces

We have the spectral decomposition of a given operator $\mathbf{A}$ given by

$$\mathbf{A} = \sum_i \lambda_i |i\rangle\langle i|,$$

with the ONB $|i\rangle$ being eigenvectors of $\mathbf{A}$ and λ_i being the eigenvalues. Similarly, a operator which is a function of $\mathbf{A}$ is given by

$$f(\mathbf{A}) = \sum_i f(\lambda_i) |i\rangle\langle i|.$$

The trace of a product of matrices is cyclic, that is

$$\mathrm{tr}[\mathbf{ABC}] = \mathrm{tr}[\mathbf{BCA}] = \mathrm{tr}[\mathbf{CBA}],$$

and we have also

$$\mathrm{tr}[\mathbf{A}|\psi\rangle\langle\psi|] = \langle\psi|\mathbf{A}|\psi\rangle.$$

Definition of density matrix

Using the spectral decomposition we defined also the density matrix as

$$\rho = \sum_i p_i |i\rangle\langle i|,$$

where the probability p_i are the eigenvalues of the density linked with the ONB $|i\rangle$.

The trace of the density matrix is $\text{tr}\rho = 1$ and it is invariant under unitary transformations $|\psi'_i\rangle = \mathbf{U}|\psi_i\rangle$. The unitary transformation of the density matrix gives, with $\mathbf{U}^\dagger \mathbf{U} = \mathbf{U}^T \mathbf{U} = \mathbf{I}$,

$$\mathbf{U}\rho\mathbf{U}^\dagger = \sum_i p_i \mathbf{U}|\psi_i\rangle\langle\psi_i|\mathbf{U}^\dagger,$$

and with the unitary transformation it is easy to show that the trace of the transformed density matrix is equal to one,

$$\text{tr} [\mathbf{U}\rho\mathbf{U}^\dagger] = \text{tr} [\mathbf{U}\mathbf{U}^\dagger\rho] = 1.$$

Discussion of exercises from last week

See jupyter-notebook from week 2

First entanglement encounter, two qubit system

We define a system that can be thought of as composed of two subsystems A and B . Each subsystem has computational basis states

$$|0\rangle_{A,B} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T \quad |1\rangle_{A,B} = \begin{bmatrix} 0 & 1 \end{bmatrix}^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry.

Computational basis

This leads to the many-body computational basis states

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = [1 \ 0 \ 0 \ 0]^T,$$

and

$$|01\rangle = |0\rangle_A \otimes |1\rangle_B = [0 \ 1 \ 0 \ 0]^T,$$

and

$$|10\rangle = |1\rangle_A \otimes |0\rangle_B = [0 \ 0 \ 1 \ 0]^T,$$

and finally

$$|11\rangle = |1\rangle_A \otimes |1\rangle_B = [0 \ 0 \ 0 \ 1]^T.$$

Bell states

The above computational basis states, which define an ONB, can in turn be used to define another ONB. As an example, consider the so-called Bell states

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle + |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix},$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}} [|00\rangle - |11\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix},$$

The next two

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} [|10\rangle + |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix},$$

and

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} [|10\rangle - |01\rangle] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

It is easy to convince oneself that these states also form an orthonormal basis.

Measurement

Measuring one of the qubits of one of the above Bell states, automatically determines, as we will see below, the state of the second qubit. To convince ourselves about this, let us assume we perform a measurement on the qubit in system A by introducing the projections with outcomes 0 or 1 as

$$P_0 = |0\rangle\langle 0|_A \otimes I_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

for the projection of the $|0\rangle$ state in system A and similarly

$$P_1 = |1\rangle\langle 1|_A \otimes I_B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

for the projection of the $|1\rangle$ state in system A .

Probability of outcome

We can then calculate the probability for the various outcomes by computing for example the probability for measuring qubit 0

$$\langle \Phi^+ | \mathbf{P}_0 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

Similarly, we obtain

$$\langle \Phi^+ | \mathbf{P}_1 | \Phi^+ \rangle = \frac{1}{2} [\langle 00 | + \langle 11 |] \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} [|00\rangle + |11\rangle] = \frac{1}{2}.$$

States after measurement

After the above measurements the system is in the states

$$|\Phi'_0\rangle = \sqrt{2} [|0\rangle\langle 0|_A \otimes I_B] |\Phi^+\rangle = |00\rangle,$$

and

$$|\Phi'_1\rangle = \sqrt{2} [|1\rangle\langle 1|_A \otimes I_B] |\Phi^+\rangle = |11\rangle.$$

We see from the last two equations that the state of the second qubit is determined even though the measurement has only taken place locally on system A .

Other states

If we on the other hand consider a state like

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B,$$

this is a pure **product** state of the single-qubit, or single-particle states, of two qubits (particles) in system A and system B , respectively. We call such a state for a **pure product state**.

Quantum states that cannot be written as a mixture of other states are called pure quantum states or just product states, while all other states are called mixed quantum states.

More on Bell states

A state like one of the Bell states (where we introduce the subscript AB to indicate that the state is composed of single states from two subsystem)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [|00\rangle_{AB} + |11\rangle_{AB}],$$

is on the other hand a mixed state and we cannot determine whether system A is in a state 0 or 1. The above state is a superposition of the states $|00\rangle_{AB}$ and $|11\rangle_{AB}$ and it is not possible to determine individual states of systems A and B , respectively.

Entanglement

We say that the state is entangled. This yields the following definition of entangled states: a pure bipartite state $|\psi\rangle_{AB}$ is entangled if it cannot be written as a product state $|\psi\rangle_A \otimes |\phi\rangle_B$ for any choice of the states $|\psi\rangle_A$ and $|\phi\rangle_B$. Otherwise we say the state is separable.

Examples of entanglement

As an example, consider an ansatz for the ground state of the helium atom with two electrons in the lowest $1s$ state (hydrogen-like orbits) and with spin $s = 1/2$ and spin projections $m_s = -1/2$ and $m_s = 1/2$. The two single-particle states are given by the tensor products of their spatial $1s$ single-particle states $|\phi_{1s}\rangle$ and their spin up or spin down spinors $|\xi_{sm_s}\rangle$. The ansatz for the ground state is given by a Slater determinant with total orbital momentum $L = l_1 + l_2 = 0$ and total spin $S = s_1 + s_2 = 0$, normally labeled as a spin-singlet state.

Ground state of helium

This ansatz for the ground state is then written as, using the compact notations

$$|\psi_i\rangle = |\phi_{1s}\rangle_i \otimes |\xi\rangle_{s_i m_{s_i}} = |1s, s, m_s\rangle_i,$$

with i being electron 1 or 2, and the tensor product of the two single-electron states as

$|1s, s, m_s\rangle_1 |1s, s, m_s\rangle_2 = |1s, s, m_s\rangle_1 \otimes |1s, s, m_s\rangle_2$, we arrive at

$$\Psi(\mathbf{r}_1, \mathbf{r}_2; s_1, s_2) = \frac{1}{\sqrt{2}} [|1s, 1/2, 1/2\rangle_1 |1s, 1/2, -1/2\rangle_2 - |1s, 1/2, -1/2\rangle_1 |1s, 1/2, 1/2\rangle_2]$$

This is also an example of a state which cannot be written out as a pure state. We call this for an entangled state as well.

Maximally entangled

A so-called maximally entangled state for a bipartite system has equal probability amplitudes

$$|\Psi\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle.$$

We call a bipartite state composed of systems A and B (these systems can be single-particle systems, or single-qubit systems representing low-lying states of complicated many-body systems) for separable if its density matrix ρ_{AB} can be written out as the tensor product of the individual density matrices ρ_A and ρ_B , that is we have for a given probability distribution p_i

$$\rho_{AB} = \sum_i p_i \rho_A(i) \otimes \rho_B(i).$$

Schmidt decomposition

If we cannot write the density matrix in this form, we say the system AB is entangled. In order to see this, we can use the so-called Schmidt decomposition, which is essentially an application of the singular-value decomposition.

Pure states and Schmidt decomposition

The Schmidt decomposition allows us to define a pure state in a bipartite Hilbert space composed of systems A and B as

$$|\psi\rangle = \sum_{i=0}^{d-1} \sigma_i |i\rangle_A |i\rangle_B,$$

where the amplitudes σ_i are real and positive and their squared values sum up to one, $\sum_i \sigma_i^2 = 1$. The states $|i\rangle_A$ and $|i\rangle_B$ form orthonormal bases for systems A and B respectively, the amplitudes λ_i are the so-called Schmidt coefficients and the Schmidt rank d is equal to the number of Schmidt coefficients and is smaller or equal to the minimum dimensionality of system A and system B , that is $d \leq \min(\dim(A), \dim(B))$.

Proof of Schmidt decomposition

The proof for the above decomposition is based on the singular-value decomposition. To see this, assume that we have two orthonormal bases sets for systems A and B , respectively. That is we have two ONBs $|i\rangle_A$ and $|j\rangle_B$. We can always construct a product state (a pure state) as

$$|\psi\rangle = \sum_{ij} c_{ij} |i\rangle_A |j\rangle_B,$$

where the coefficients c_{ij} are the overlap coefficients which belong to a matrix $\mathbf{C}$.

Further parts of proof

If we now assume that the dimensionalities of the two subsystems A and B are the same d , we can always rewrite the matrix $\mathbf{C}$ in terms of a singular-value decomposition with unitary/orthogonal matrices $\mathbf{U}$ and $\mathbf{V}$ of dimension $d \times d$ and a matrix $\mathbf{\Sigma}$ which contains the (diagonal) singular values $\sigma_0 \leq \sigma_1 \leq \dots 0$ as

$$\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\dagger.$$

SVD parts in proof

This means we can rewrite the coefficients c_{ij} in terms of the singular-value decomposition

$$c_{ij} = \sum_k u_{ik} \sigma_k v_{kj},$$

and inserting this in the definition of the pure state $|\psi\rangle$ we have

$$|\psi\rangle = \sum_{ij} \left(\sum_k u_{ik} \sigma_k v_{kj} \right) |i\rangle_A |j\rangle_B.$$

Slight rewrite

We rewrite the last equation as

$$|\psi\rangle = \sum_k \sigma_k \left(\sum_i u_{ik} |i\rangle_A \right) \otimes \left(\sum_j v_{kj} |j\rangle_B \right),$$

which we identify simply as, since the matrices $\mathbf{U}$ and $\mathbf{V}$ represent unitary transformations,

$$|\psi\rangle = \sum_k \sigma_k |k\rangle_A |k\rangle_B.$$

Different dimensionalities

It is straight forward to prove this relation in case systems A and B have different dimensionalities. Once we know the Schmidt decomposition of a state, we can immediately say whether it is entangled or not. If a state ψ is entangled, then its Schmidt decomposition has more than one term. Stated differently, the state is entangled if the so-called Schmidt rank is greater than one. There is another important property of the Schmidt decomposition which is related to the properties of the density matrices and their trace operations and the entropies. In order to introduce these concepts let us look at the two-qubit Hamiltonian described here.

Entropies and density matrices

Note: more details on whiteboard. This material will be added later

Shannon information entropy

We start our discussions with the classical information entropy, or just Shannon entropy, before we move over to a quantum mechanical way to define the entropy based on the density matrices discussed earlier.

We define a set of random variables $X = \{x_0, x_1, \dots, x_{n-1}\}$ with probability for an outcome $x \in X$ given by $p_X(x)$, the information entropy is defined as

$$S = - \sum_{x \in X} p_X(x) \log_2 p_X(x).$$

Von Neumann entropy

$$S = -\mathrm{Tr}[\rho \log_2 \rho].$$

Two-qubit system and calculation of density matrices and exercise

This part is best seen using the jupyter-notebook.

The system we discuss here is a continuation of the two qubit example from week 2.

This system can be thought of as composed of two subsystems A and B . Each subsystem has computational basis states

$$|0\rangle_{A,B} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T \quad |1\rangle_{A,B} = \begin{bmatrix} 0 & 1 \end{bmatrix}^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry. This leads to the many-body computational basis states

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T,$$

and

$$|01\rangle = |0\rangle_A \otimes |1\rangle_B = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}^T,$$

and

$$|10\rangle = |1\rangle_A \otimes |0\rangle_B = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}^T,$$

Use the Hamiltonian for the two-qubit example to find the eigenpairs as functions of the interaction strength λ and study the final eigenvectors as functions of the admixture of the original basis states. Discuss the results as functions of the parameter λ and compute the von Neumann entropy and discuss the results. You will need to calculate the entropy of the subsystems A or B .

The next lecture

In our next lecture, we will discuss

1. Reminder and review of entropy and entanglement
2. Gates and circuits and how to perform operations on states

Reading: Chapters 2.1-2.11 of Hundt's text